

Make A Battery-Operated Portable Temperature Measurement System

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This project presents a K-type thermocouple-based temperature measurement system using MSP430G2553 microcontroller (MCU).

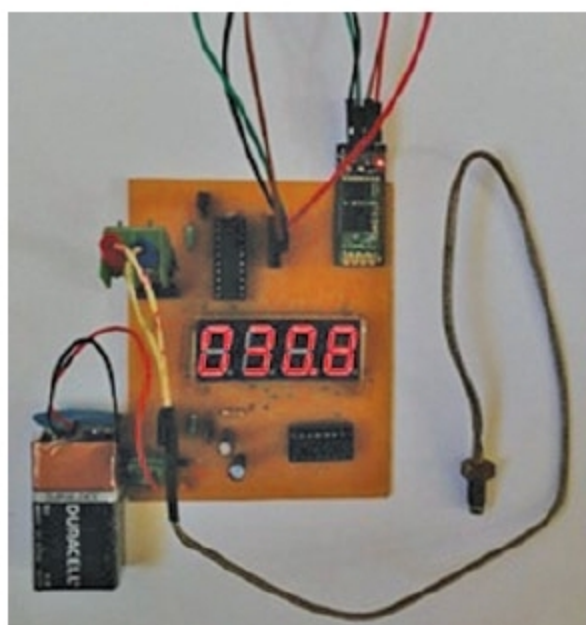


Fig. 1: Author's prototype

The MCU reads sensor data and updates it on a four-digit, seven-segment display, and also sends it to a smartphone through Bluetooth module. The author's prototype is shown in Fig. 1 and its block diagram in Fig. 2.

Temperature measurement is vital in various industrial and weather-

monitoring applications. Resistance temperature detectors (RTDs), thermistors and thermocouples are commonly used as sensors for temperature measurement in industrial applications. K-type thermocouples are widely used in industrial applications as these work well in rugged envi-

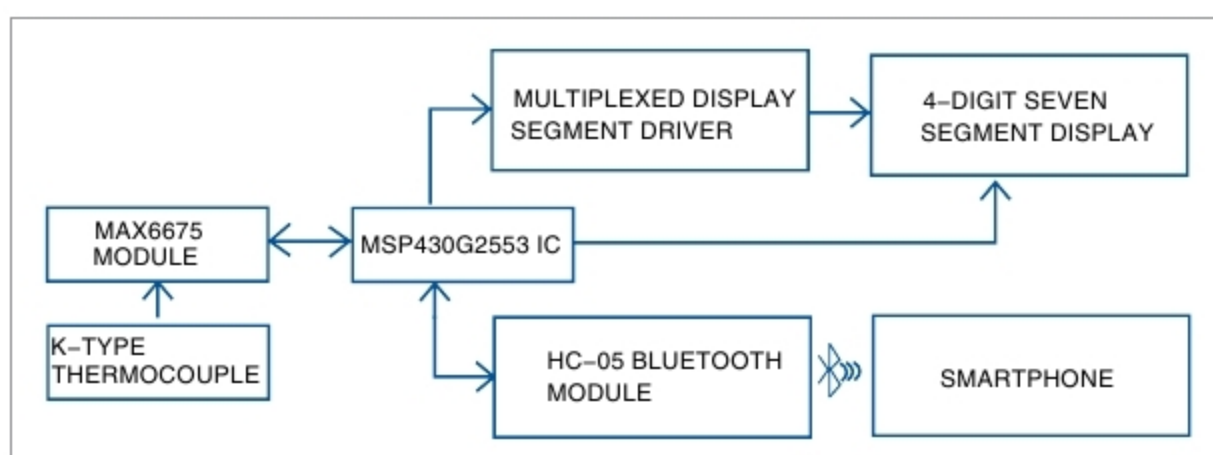


Fig. 2: Block diagram of the temperature measurement system

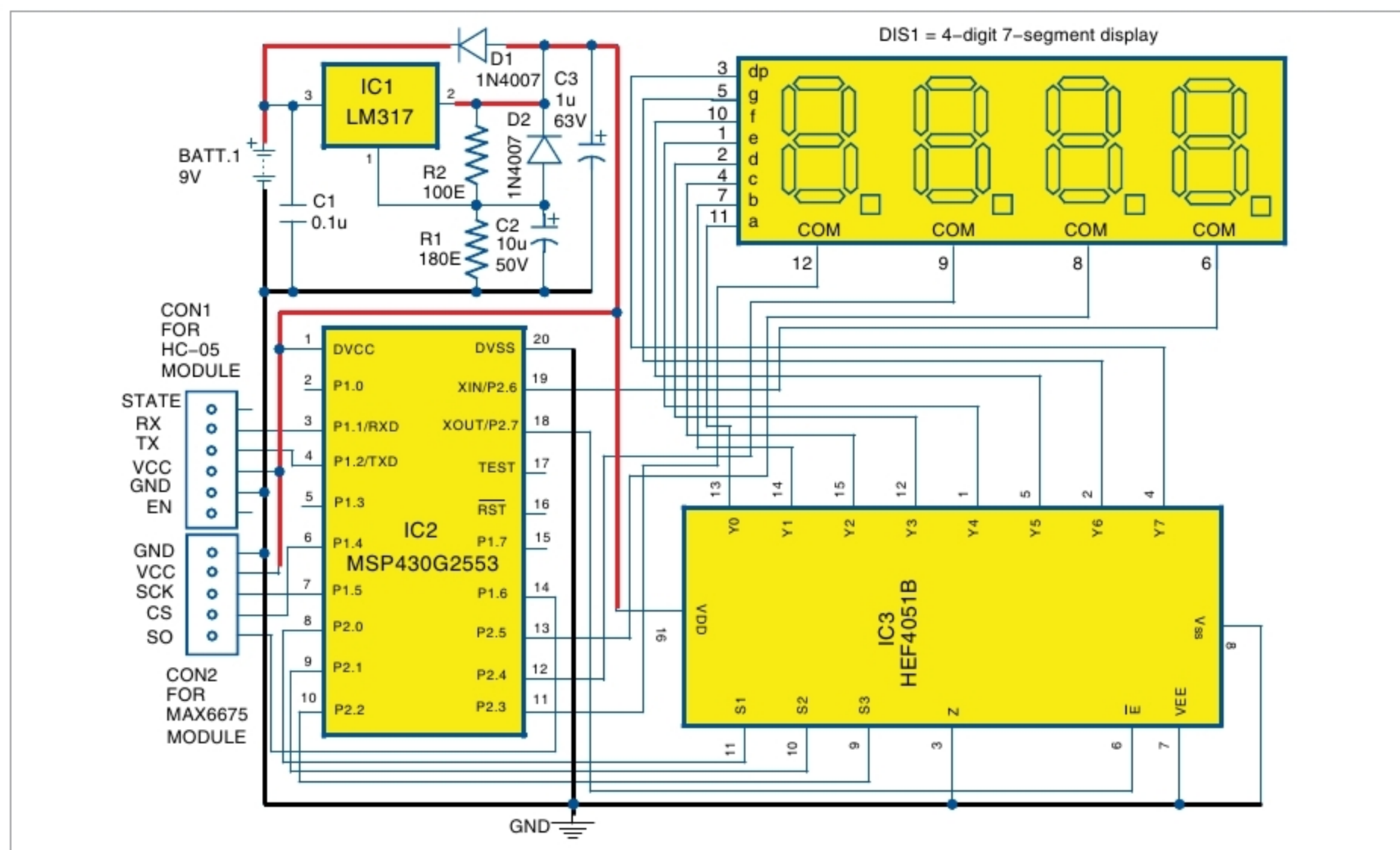


Fig. 3: Circuit diagram of the temperature measurement system